

KaraoScore

Team:

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Github Repo: <https://github.com/AnthonyLi88/KaraoScore>

License: GNU GPLv3

Overall Vision:

Our vision is to make a web app that gamifies your singing. You submit either an audio file or link to a song and another of the instrumentals (technically can be used with any audio file if you wanted), and you will be put into a virtual karaoke room where you sing into a microphone. After you've finished singing, your singing will be combined with the instrumental track and compared to the original track.

Stacks

- **Frontend:** React or Vue.js.
- **Styling:** Tailwind CSS.
- **Backend/Database:** Supabase or Firebase
 - **Libraries:**
 - YT-DLP
 - FFMPEG
 - Wavesurfer.js
 - MediaRecorder API
 - Pitchfinder.js

Semester Goals:

1. Create a functional MVP (or minimal viable product): Basically, make a bare-bones application with the basic “gameplay loop” in full functionality.
2. Open source contribution: Release the project as open-source to allow others to collaborate
3. Learn and feel comfortable working with React Native
4. Make Anthony proud!!

Timeline Goals

September

- Environment setup. Initialize Electron with React/Vue.
- Research on audio extraction methods, audio web processing, and audio libraries for our website.
- Research on copyright laws involving music streaming
- Conceptualize a core audio pipeline

October

- Put together an audio extraction and then syncing algorithm that will be the core part of the project.
- Explore frontend designing.
- Experiment with making it more user friendly, rather than assuming the user will use it correctly

November

- Focus on polishing major bugs
- Calibrating the scoring system of the website's core activity

December

- Have MVP of the basic program vision that people can enjoy

Individual Roles

Tony Jiang

- Initialize Electron with React/Vue.
- Develop the audio processing logic behind the app.
- Backend designing
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Anthony Li

- Environment setup.
- Utilize yt-dlp library for audio extraction of music.
- Frontend designing
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